Tool Support for Causal Reasoning via String Diagram Surgery

Fabio Zanasi



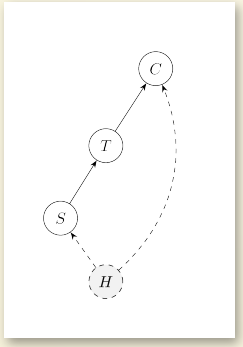
Joint work with Sifei Li

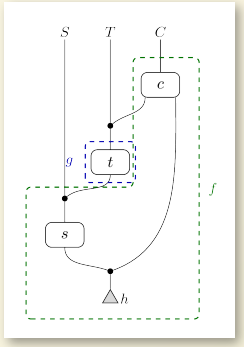
ICTAC 2026

Overview

Scope: **causal** and **counterfactual** reasoning in **Bayesian networks**.

Categorical semantics via
string diagram surgery



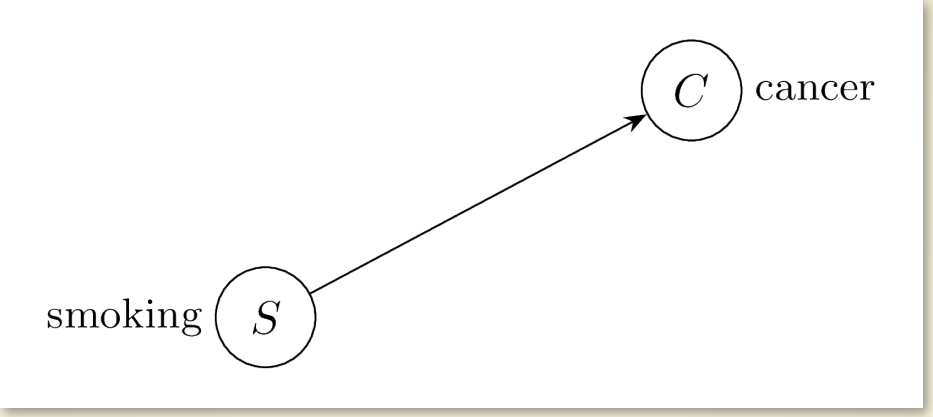


In this work: **software implementation** of the categorical approach, **assessment** and **comparison** with existing tools.

Part I
Causal Identifiability

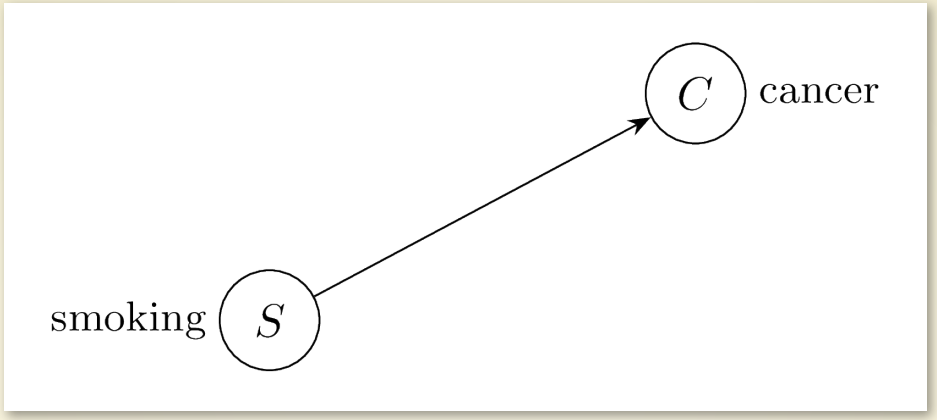
The Smoking Scenario

Does smoking cause cancer?



The Smoking Scenario

Does smoking cause cancer?



Observational data

$P(S, C)$

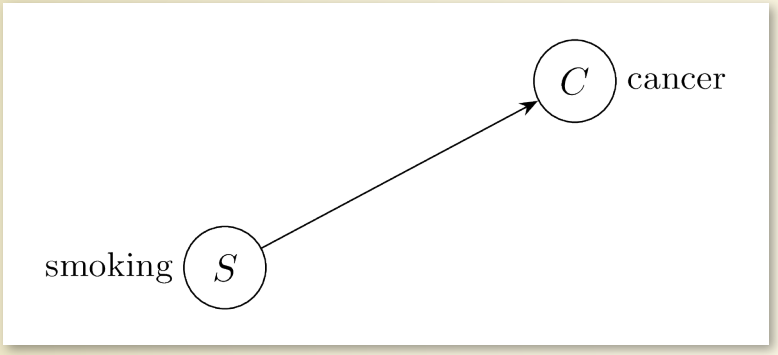
P(cancer | smoke) = 0.68

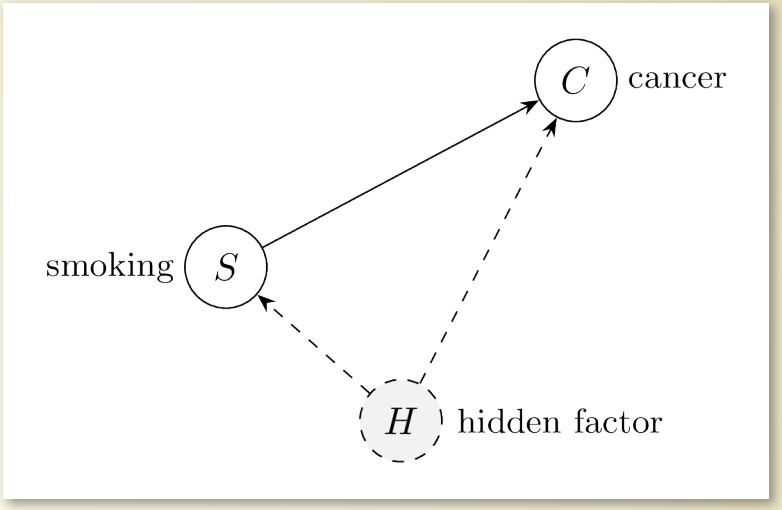
P(cancer | no smoke) = 0.19

The scientist says: yes.

The Smoking Scenario

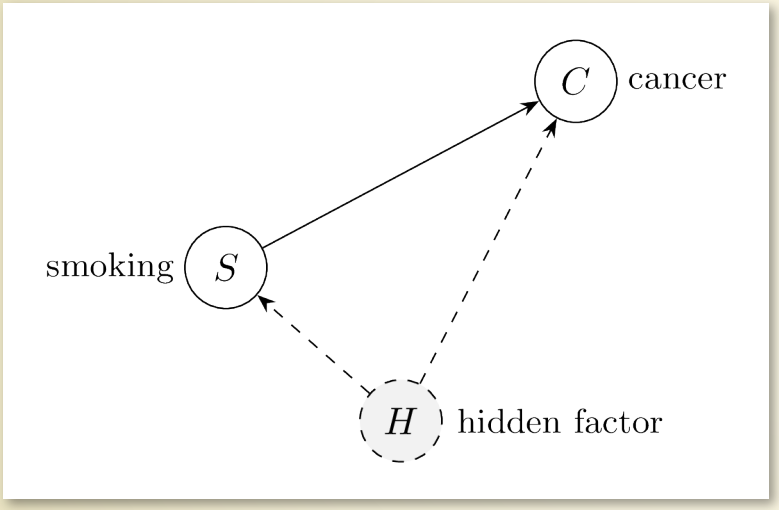
Does smoking cause cancer?





The tobacco company says: maybe not.
A hidden factor (say, a genetic trait) could cause both the craving and the disease.

Three questions about smoking

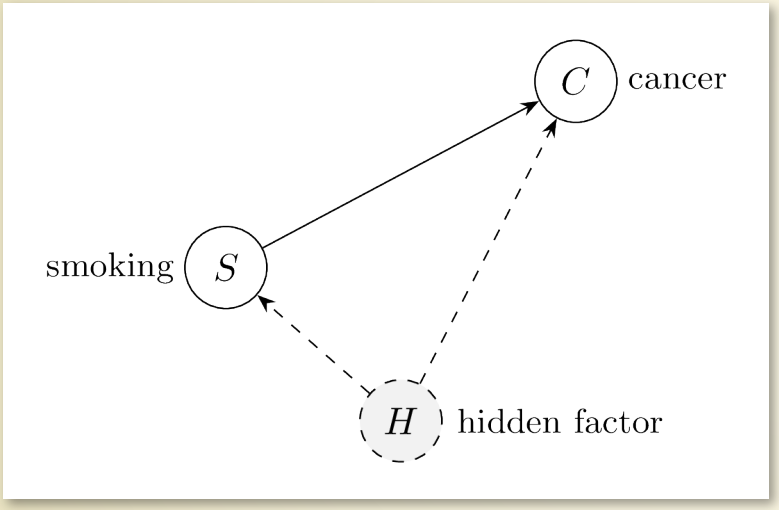


Association

Do smokers get cancer more often?

$$P(C \mid S)$$

Three questions about smoking



Association

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$$P(C \mid S)$$

Intervention

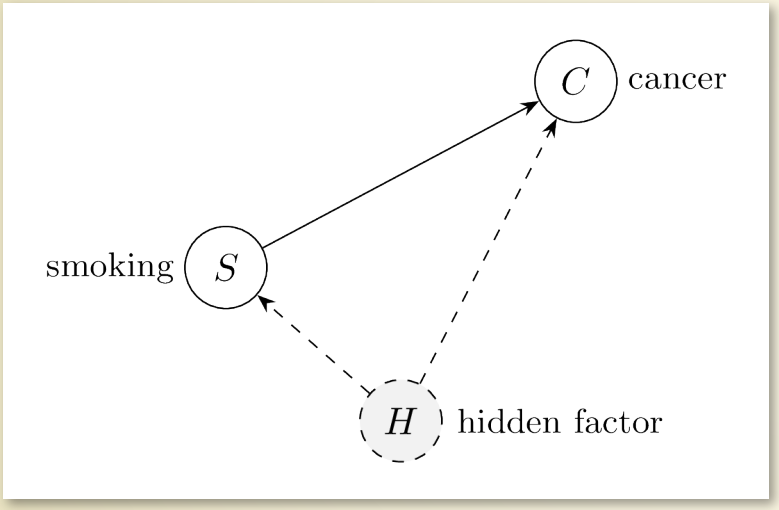
Does smoking cause cancer?

$$P(C \mid \mathrm{do}(S))$$

Pearl's do-operator

a randomised trial would tell … but is not allowed

Three questions about smoking



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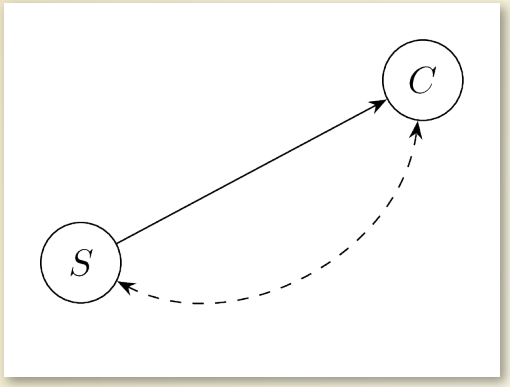
Counterfactual

Bob smoked and got cancer: would he have got cancer had he not smoked?

$$P(C_{\neg s} \mid S{=}s,\ C{=}c)$$

Identifiability

Can the query be computed from the observational distribution alone?

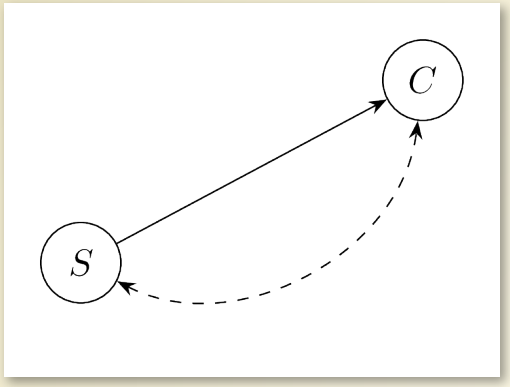


✗ not identifiable

two models can agree on P(S, C) and disagree on P(C | do(S))

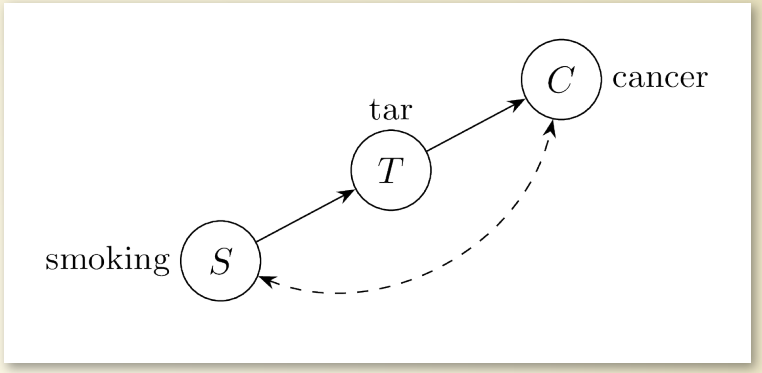
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✓ **identifiable**

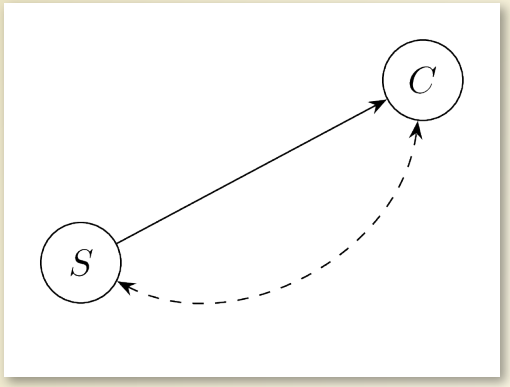
$$P(c \mid \mathrm{do}(s)) = \sum_t P(t \mid s) \sum_{s'} P(c \mid t, s') P(s')$$

(Pearl's front-door formula)

here: P(cancer | do(smoke)) = 0.54
P(cancer | do(no smoke)) = 0.25

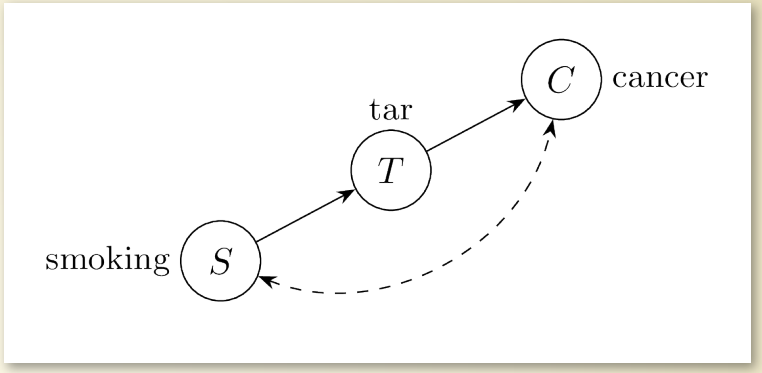
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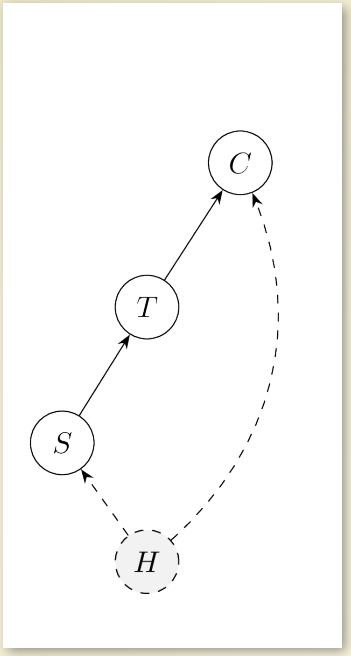
here: P(cancer | do(smoke)) = 0.54
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Complete algorithms

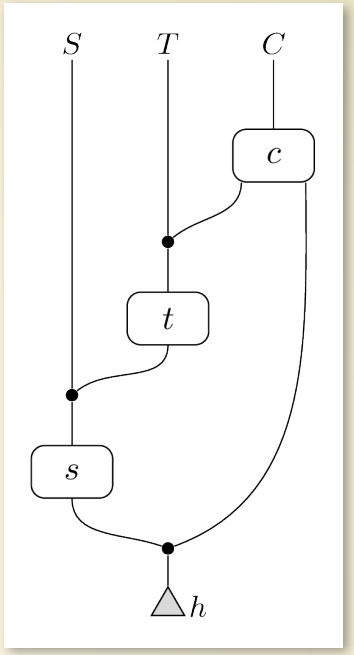
- ID (Shpitser–Pearl 2006) for interventions; ID* / IDC* (Shpitser–Pearl 2008) for counterfactuals.
- Implementations: R cfid (Tikka 2023), Python y0 (Hoyt et al. 2025): graph algorithms plus symbolic bookkeeping.

Part II
String Diagram Surgery

Bayesian networks as string diagrams

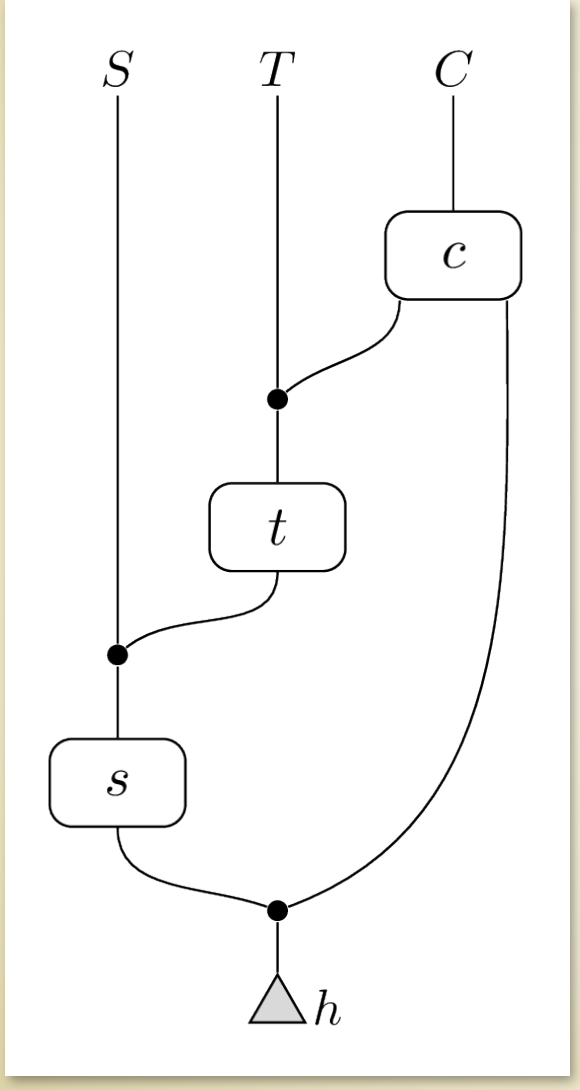






- Morphism of a symmetric monoidal category
- Algebraic object (proofs/defs by induction)
- Separation of syntax and semantics
- Compositionality
- Axiomatic reasoning

Bayesian networks as string diagrams



Read bottom to top:

- wires are variables; boxes are conditional distributions P(T | S), P(C | T, H), …
- ▲ is a distribution with no inputs (a state); • copies a value.
- Observable variables = wires on the interface

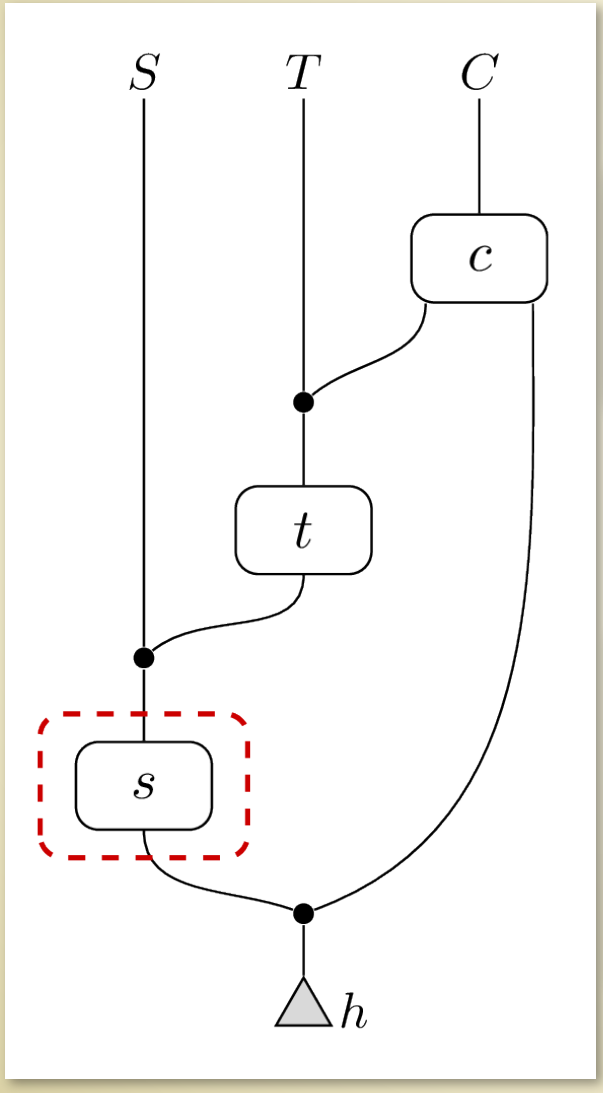
Categorically:

It's a morphism

$$\omega : I \to S \otimes T \otimes C$$

in the monoidal category **FStoch** of finite sets and stochastic matrices.

Interventions as diagram surgery



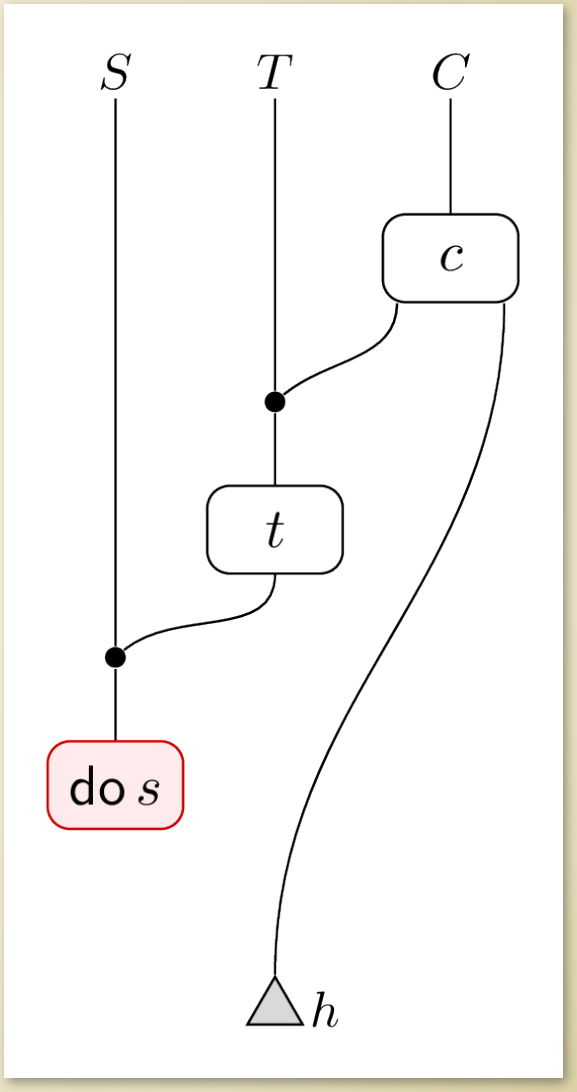
cut

Pearl's do-operator = Local rewrite of the diagram

Endofunctor in **FStoch**

Identifiability = a certain diagram shape (*comb*)

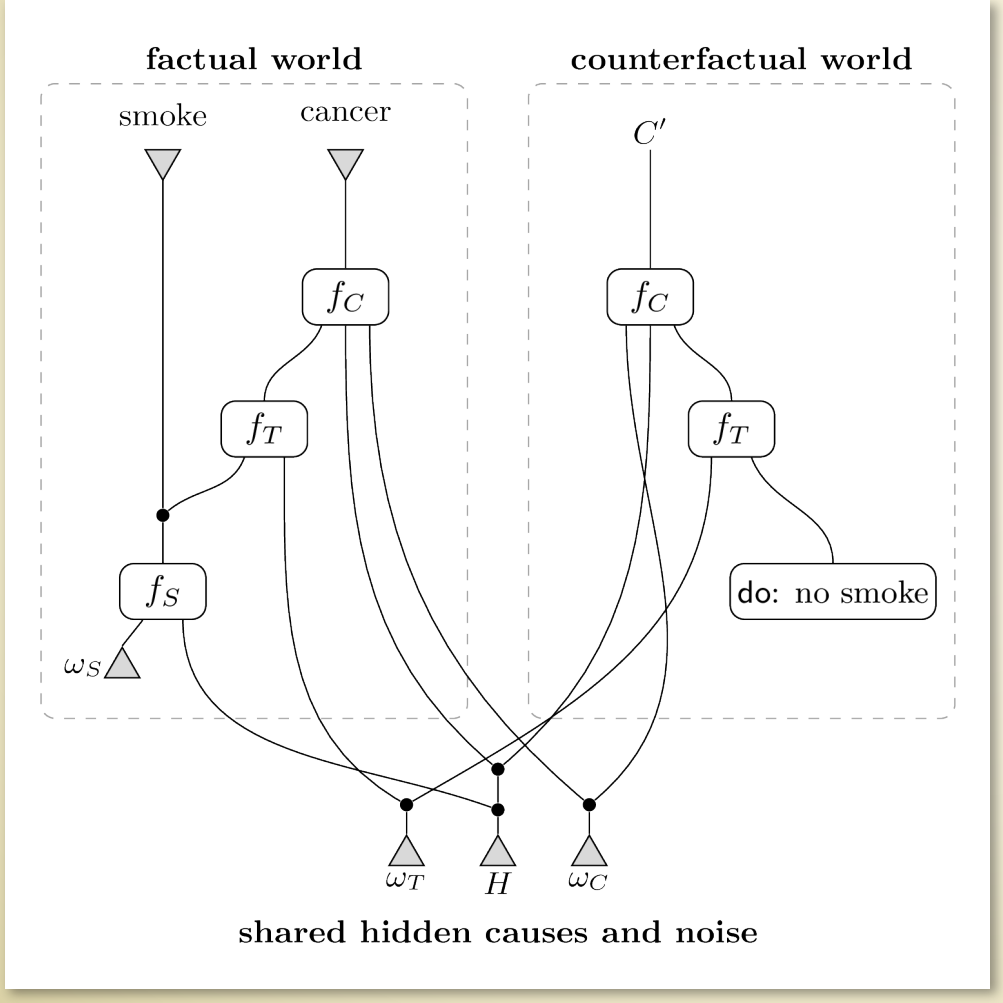
Computing the intervention = *comb disintegration*



Jacobs, Kissinger, Z. — FoSSaCS 2019, MSCS 2021

Counterfactuals via twin networks (Sketch)

Methodology: put the two worlds side by side, sharing hidden causes and noise



$$P(C_{\neg s} \mid S{=}s,\ C{=}c)$$

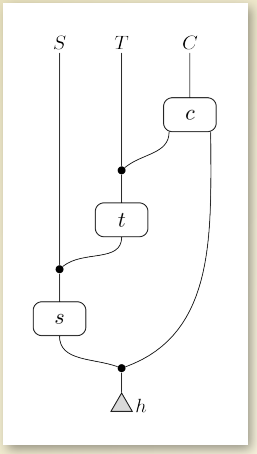
Factual world: Bob smoked and got cancer: the evidence, drawn as effects ▽.

Counterfactual world: the surgery do(no smoke) replaces the mechanism of S.

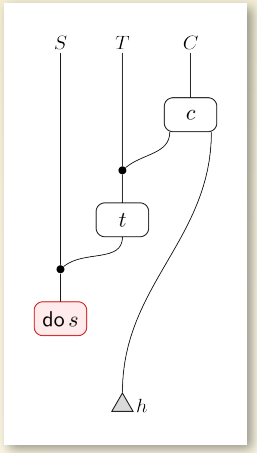
Shared: the hidden factor H and the noise, so both worlds concern the same Bob.

Query: the output wire C′, conditioned on the evidence.

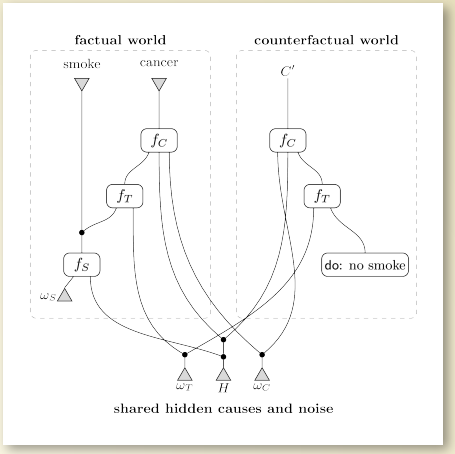
Summing up string diagram surgery



model



intervention



counterfactual

Common language Model, intervention, counterfactual and estimand are all string diagrams in a symmetric monoidal category.

Equational reasoning Surgery, simplification and identification are rewrites, sound by the laws of the category.

Compositionality Models are both combinatorial and syntactic objects, whose properties may be defined and proved by structural induction.

Theory: Jacobs–Kissinger–Z. 2019/2021, Lorenz–Tull 2023, Friend et al. 2025 …
but no software implementation.

Part III
Implementing String Diagram Surgery

Questions we want to answer

Feasibility

Can the categorical semantics be turned into a working tool for causal reasoning, and what does it take in practice?

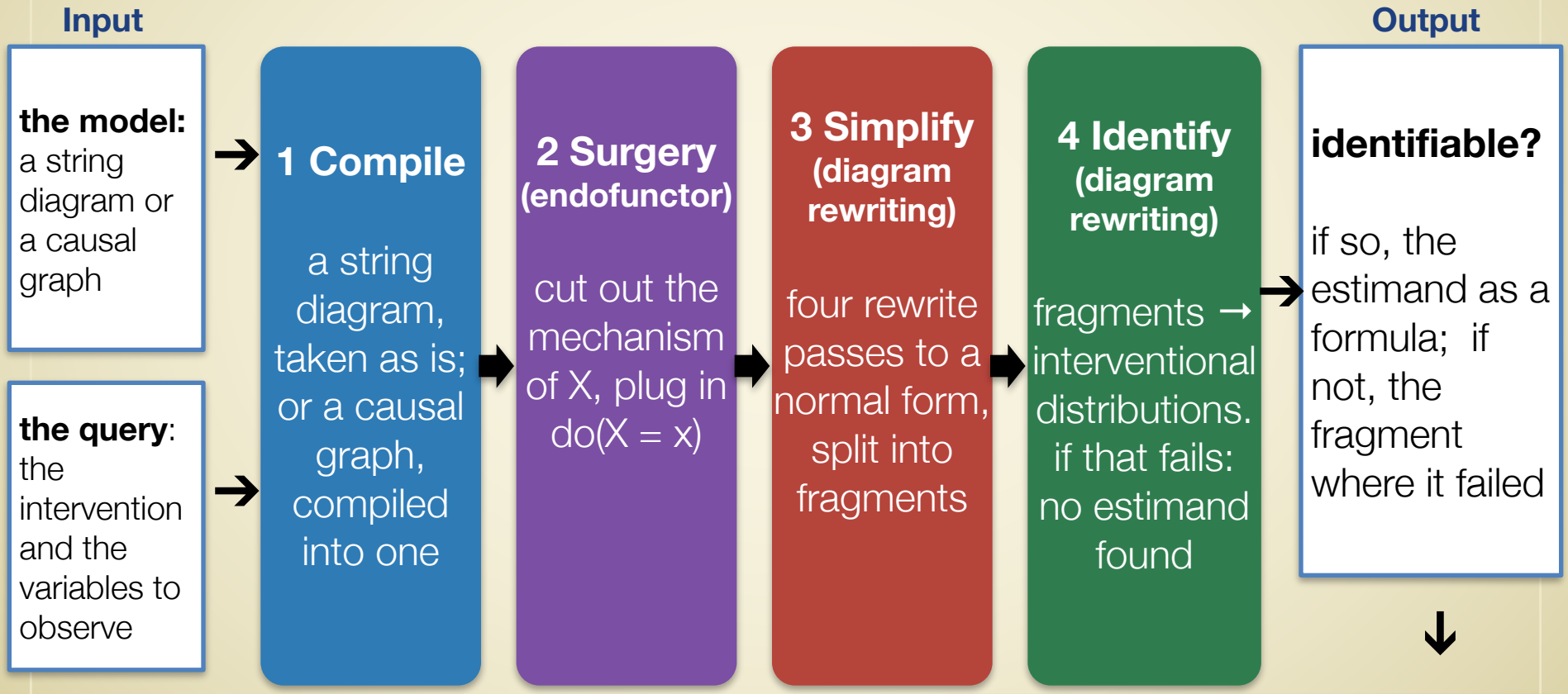
Comparison

How does it fare against the established combinatorial and symbolic tools (R cfid, Python y0, pyAgrum), in correctness, runtime and transparency?

The tool

`github.com/GrayratA/idcf-string-diagram-toolkit`

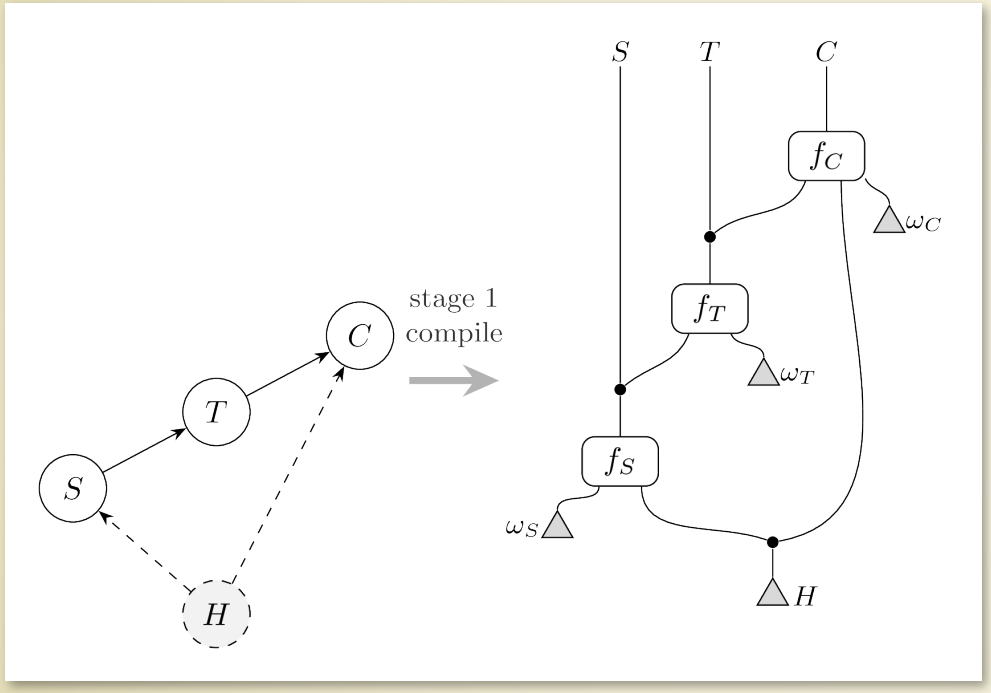
A Julia prototype, based on the Catlab library



every stage is exportable to Chyp for visual inspection

Scoped extension: if the model has a comb shape, comb disintegration also computes the estimand numerically.

Worked example: the smoking model



Input

the model as a string diagram

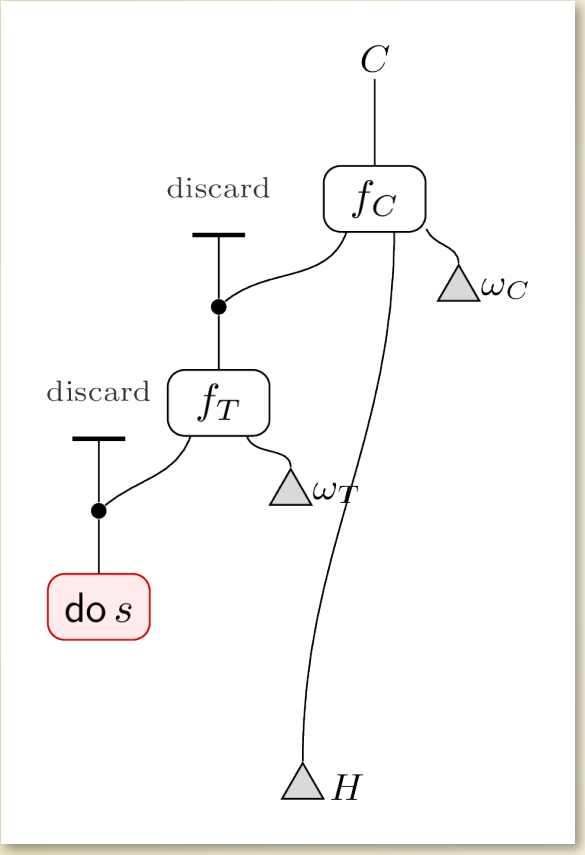
OR

as a causal graph with directed edges and hidden common causes, which stage 1 compiles into the same diagram.

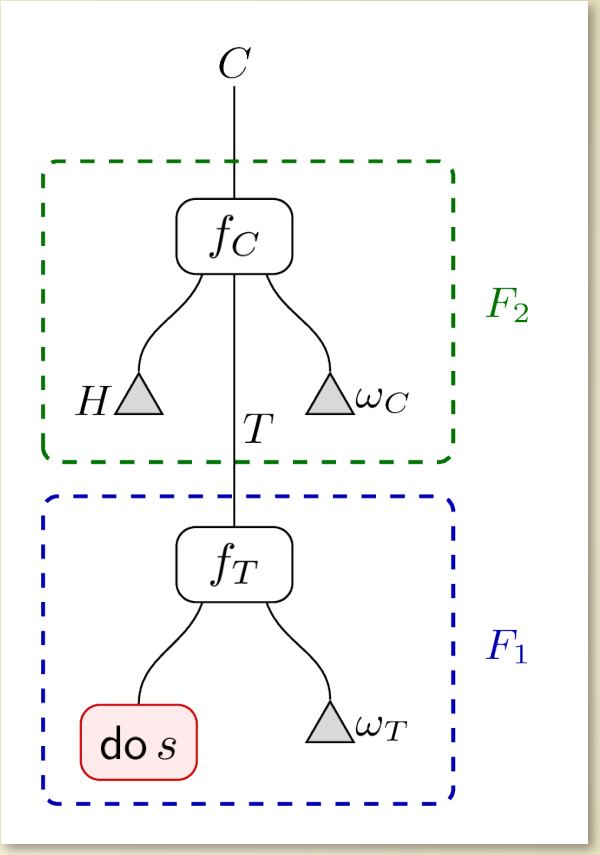
Query

$$P(C \mid \mathrm{do}(S{=}s))$$

Stages 2–3: surgery, then simplification

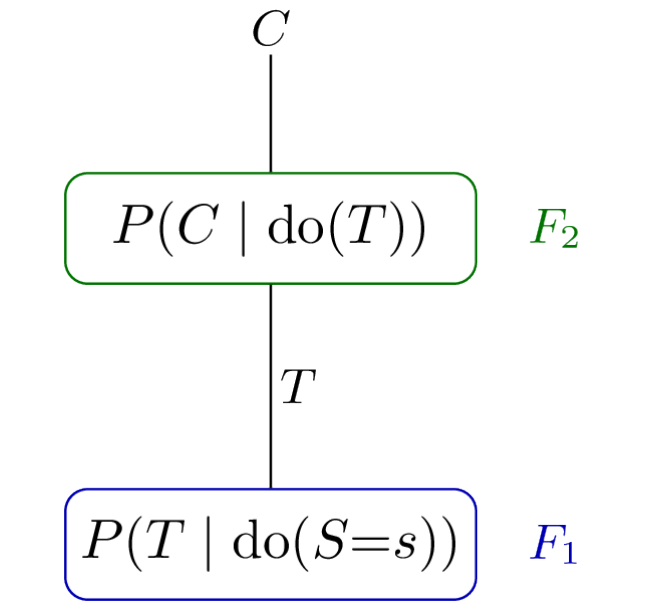


Stage 2: cut out f_S and plug in the state **do** s; outputs that are not asked for are discarded (= postcomposed with counit string diagram).



Stage 3: four diagram rewrite passes (prune, separate, merge, absorb) give a canonical form, split into R-fragments F1, F2.

Stages 4: identification



Each fragment denotes an interventional probability.

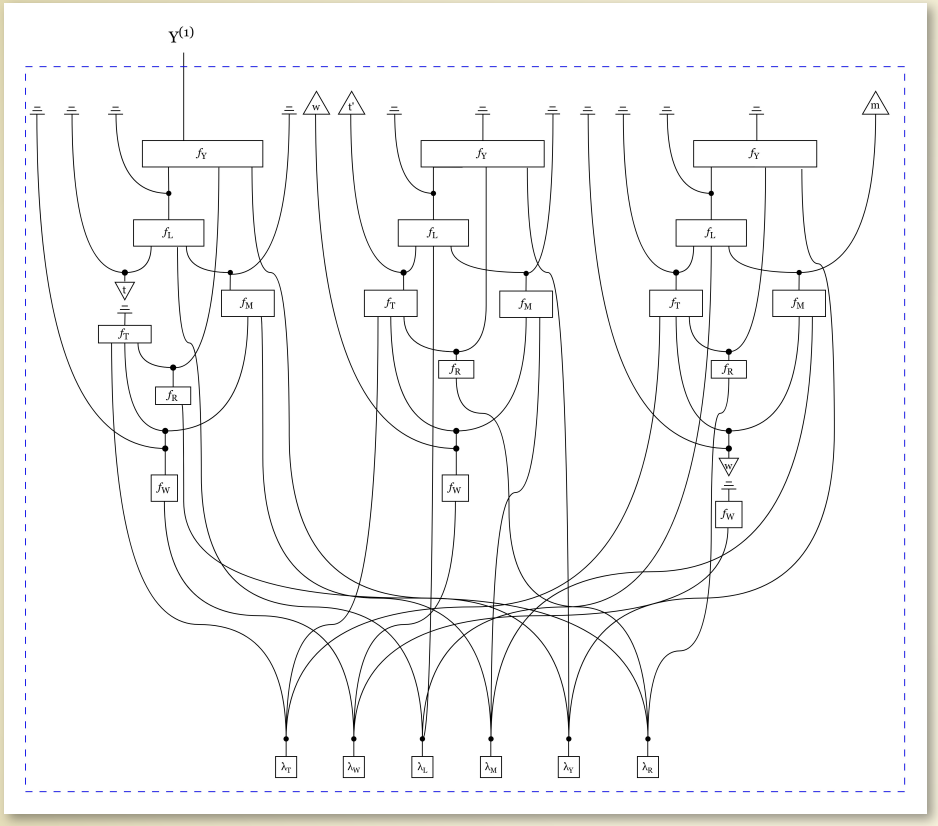
Compose the fragments back along T: the resulting diagram denotes the estimand, which is the tool's output (on this model it corresponds to the front-door formula)

$$P(c \mid \mathrm{do}(s)) \;=\; \textstyle\sum_t P\big(t \mid \mathrm{do}(s)\big)\; P\big(c \mid \mathrm{do}(t)\big)$$

$$\Rightarrow\; P(c \mid \mathrm{do}(s)) \;=\; \textstyle\sum_t P(t \mid s) \sum_{s'} P(c \mid t, s')\, P(s')$$

When identification fails: a fragment does not denote an interventional probability, because an exogenous source it shares with the rest of the diagram cannot be absorbed into it. The tool reports that fragment and that source.

Counterfactuals: the same pipeline (sketch)



Input

the commute model: departure time T and performance Y share a hidden cause

Query

she left at t′ under weather w and took mode m: how would she have performed, had she left at t?

$$P\big(Y_t \mid T{=}t',\ W{=}w,\ M_w{=}m\big)$$

Stage 2 builds one world per clause of the query, with the noise sources shared across worlds. Stages 3–4 run as before.

A noise source shared across worlds that no fragment absorbs: no estimand found.

Output: the estimand

$$\sum_{l} P\big(T{=}t', Y \mid \mathrm{do}(W{=}w), \mathrm{do}(L{=}l)\big) \cdot P\big(L{=}l \mid \mathrm{do}(M{=}m, T{=}t)\big)$$

$$\cdot\ P\big(M{=}m \mid \mathrm{do}(W{=}w)\big) \cdot P(W{=}w)$$

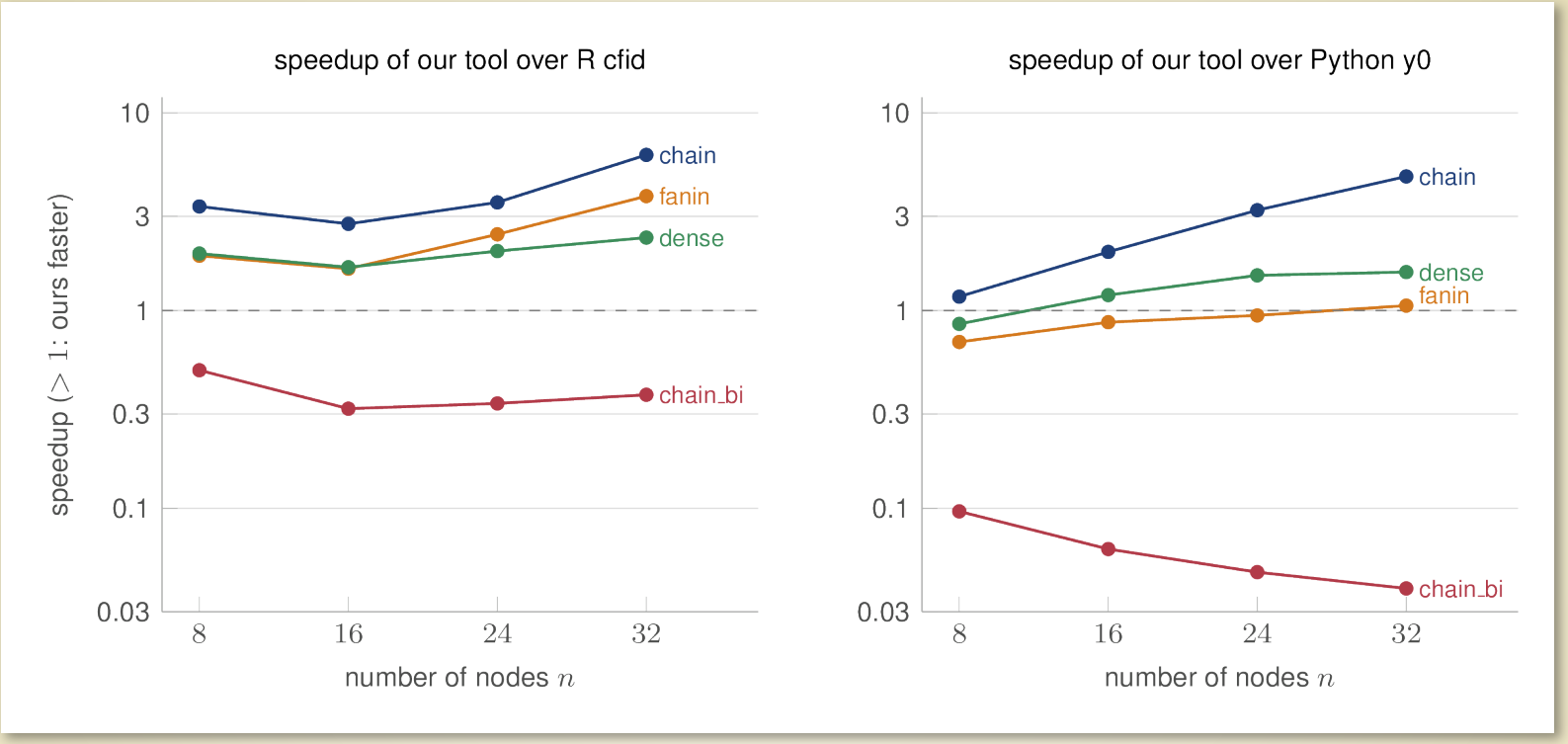
Part IV
Assessment

Correctness

1 860	98.5 %	28
test cases	agreement with cfid	disagreements
generated at random	the reference R implementation	all checked by hand

- ✓ every estimand we return is correct: sound by construction
- ✓ we largely agree with cfid
- ✗ the disagreements: in 27 cases cfid is wrong; in 1 case we miss an estimand that exists (one rewrite rule still to add)

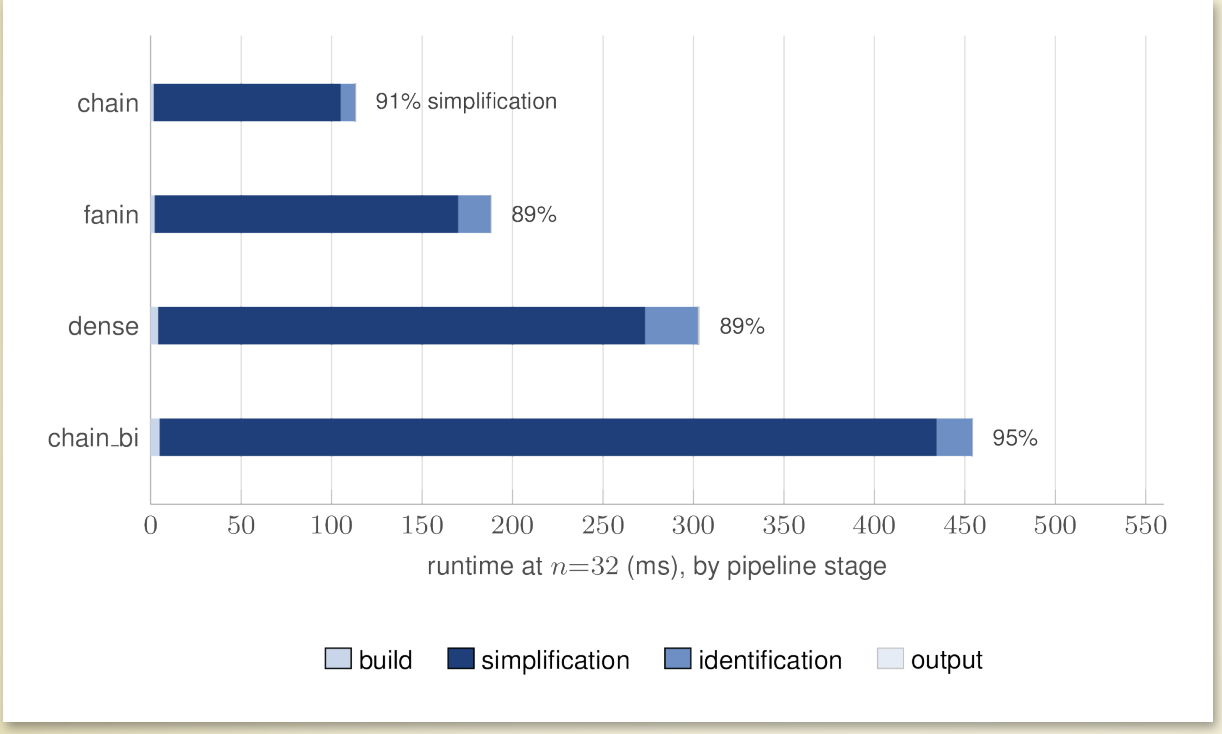
Runtime



Each colour is a family of random models of growing size; in chain_bi every variable is confounded. Above the dashed line our tool is faster than the existing one (cfid on the left, y0 on the right).

We are faster on the first three families, and the gap grows with the size of the model. On the fully confounded family we are slower: there the simplification stage blows up.

Where does the time go?



Simplification takes about 90 % of the runtime; identification (stage 4) is light.

cfid instead does the main computation inside one identifiable call.

The categorical method trades a heavy simplification phase for a light identification phase: a win when simplification shrinks the problem, a loss under dense confounding. Fix: incremental rewriting, better wiring-diagram data structures.

Conclusions

Feasible. The categorical semantics of causal reasoning can be run: the full ID-CF pipeline as rewrites on wiring diagrams, every stage inspectable.

Competitive. 98.5 % agreement with cfid and no false positives; faster on sparse and moderately dense models.

Limitations. Finite discrete models only, no causal discovery; simplification bottleneck under dense confounding; Chyp visualisation does not scale; not yet formally verified.

Next. Incremental rewriting and better data structures; an interactive visualiser; formal verification; richer probabilistic toolchains (EfProb) and links to causal representation learning.

`github.com/GrayratA/idcf-string-diagram-toolkit`

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